

100 Days of Azure DevOps

Day 72 Handout — A workspace is a database for telemetry — TimeGenerated is part of every honest question

8 - Monitoring & Observability · Personal lab only · Educational content · Not a sales pitch

Architecture A — workspace, table, query (KQL is not Excel)

Piece	What it is	Lab table
Workspace	The store. Retention and access live here	law-lab from Day 71
Table	AzureActivity, Heartbeat, KV audit, App traces — schema differs	AzureActivity is friendly even when Heartbeat is empty (no VMs)
KQL	where / summarize / join / top. SQL's cousin in the cloud	Always filter TimeGenerated. Unbounded queries wait and pay
Saved query	Workspace notebook so tomorrow-you does not reinvent the where	Save one query by name
Empty result	A result. New labs are quiet	Do not fake a graph. Enable Day 71 diagnostics first

Architecture B — pick queries that match what you actually ingested

Question	Default KQL	Wrong table
Who changed ARM today?	AzureActivity where TimeGenerated > ago(1d) summarize count() by OperationNameValue	Heartbeat — that is agent/VM pulse
Are agents alive?	Heartbeat summarize Last=max(TimeGenerated) by Computer	AzureActivity — control plane is not a VM ping
Top noisy operations	top 10 by count_	Export to Excel to 'just filter'
App 500s	App Insights requests/traces (Day 73) or AppService logs if diagnosed	AzureActivity Put WebApps

Architecture C — pitfalls the cousin judges you for

Pitfall	What you see	Fix
No TimeGenerated filter	Query spins; bill ticks	where TimeGenerated > ago(1d) first
Screenshot of 10k rows	Nobody summarizes	summarize + top 10
Heartbeat on a VM-less sub	Empty, assumed 'broken KQL'	Use AzureActivity; empty Heartbeat is honest
Query only in chat	Lost tomorrow	Save in the workspace
Wrong workspace	Tables you expected from Day 71 missing	Picker in Logs blade = law-lab, not a work default

One-liner to remember

KQL is SQL's cousin who lives in the cloud and judges your where-clauses. Ask a precise question with TimeGenerated. Excel is not a Log Analytics strategy.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal law-lab. If the workspace is empty, send a diagnostic from Day 71 before inventing tables.

- 1 Portal → Log Analytics workspace law-lab → Logs. Confirm you are not in an employer workspace.

- 2 Run query 1: AzureActivity | where TimeGenerated > ago(1d) | summarize count() by OperationNameValue | top 10 by count_.
- 3 Run query 2: a where on a specific OperationNameValue you saw, or AzureActivity | take 10 if the lab is quiet — still a result.
- 4 Run query 3: Heartbeat | take 5. If empty, write 'no VMs onboarded' — do not pretend. Optionally query the Key Vault audit table if Day 71 sent AuditEvent.
- 5 Save one query with a name (e.g. lab-azureactivity-topops). Screenshot results without tenant gossip.
- 6 Write docs/kql-day72.md: the three queries, which tables were empty, and why TimeGenerated is mandatory.

Starter KQL — time bound, summarize, top

```
AzureActivity
| where TimeGenerated > ago(1d)
| summarize count() by OperationNameValue
| top 10 by count_

AzureActivity
| where TimeGenerated > ago(1d)
| where OperationNameValue has "Microsoft.Resources/subscriptions/resourcegroups"
| project TimeGenerated, Caller, OperationNameValue, ActivityStatusValue

Heartbeat
| where TimeGenerated > ago(1h)
| summarize LastSeen=max(TimeGenerated) by Computer
```

Done checklist

- Three queries ran against the personal workspace
- One query saved
- Can explain why Heartbeat may be empty and AzureActivity still useful
- Every query used a TimeGenerated filter (or take with a reason)

Tomorrow — Day 73

Application Insights — a GoPro on the app, connection string not in git